2026년 9월 3일 목요일 오후 5:56

Loaded 101 rows

x_dot numerical (t=0.00): 0 analytic: 0

x_dot numerical (t=1.00): 0 analytic: 0

Max absolute error: 0

Mean absolute error: 0

x_dot_numerical	101
x_dot_analytic	101
absolute_errors	101
max_absolute_error	1
mean_absolute_error	1

x_dot_numerical = np.gradient(time_values)

x_dot_analytic = A·(π/T)·cos(πt/T)

absolute_errors = np.abs(x_dot_numerical - x_dot_analytic)

```
print(f">x_dot numerical (t=0.00): {x_dot_numerical[0]} analytic: {x_dot_analytic[0]}")
print(f">x_dot numerical (t=1.00): {x_dot_numerical[50]} analytic: {x_dot_analytic[50]}")
Print(f"Max absolute error: {np.max(absolute_errors)}")
print(f"Mean absolute error: {np.mean(absolute_errors)}")
```

오류 수정

>x_dot_numerical = np.gradient(x_values, time_values)

np.gradient에선 인자가 두개 필요 dx/dt